Ball and Beam Experiment Report

Modeling, calibration, control implementation, and April 11 telemetry results

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Chapter 1

Introduction

This document provides a standalone engineering report system for the ball and beam experiment contained in this repository. The report combines the canonical modeling record in `modeling.md`, calibration evidence from the April 7, 2026 beam-angle campaign, and the staircase telemetry run stored in `telemetry_logs/20260411_142117/telemetry.csv`.

The report structure is intentionally self-contained. All LaTeX sources, bibliography files, generated figures, generated tables, and build outputs live inside `report/`. The analysis pipeline reads the raw CSV logs directly and exports report-ready plots and tabular summaries without depending on any other repository.

1.1 Purpose

The main goals are:

- capture the first-principles ball and beam model used for the project,
- document the implemented sensing, calibration, and cascaded controller in the firmware,
- analyze the available calibration and staircase telemetry logs directly from raw CSV data, and
- provide a maintainable report build flow that works from the terminal and in VS Code with LaTeX Workshop.

1.2 Data Sources

The report uses the following repository-local sources as its primary experiment record:

- `modeling.md` for the mathematical derivation, sign conventions, and implementation record,
- `src/main.cpp` for the shipped controller constants and mode logic,
- `telemetry_logs/20260411_142117/telemetry.csv` for the closed-loop staircase experiment, and
- `calibration_logs/combined_20260407_forward_reverse/combined_cleaned_calibration.csv` for beam-angle calibration hysteresis characterization.

1.3 Report Roadmap

Chapter 2 summarizes the physical apparatus, coordinates, and measured parameters. Chapter 3 reproduces the nonlinear derivation from the canonical model record. Chapter 4 specializes the model around the horizontal operating point for control design. Chapter 5 documents the firmware implementation actually used in the hardware run. Chapter 6 analyzes the beam-angle calibration data. Chapter 7 discusses the April 11 telemetry run using plots and performance metrics generated directly from the raw CSV logs. Chapter 8 closes with key findings and immediate next steps.

Chapter 2

System Description

2.1 Physical Arrangement

The apparatus is a single-axis ball and beam system. A ping-pong ball rolls along a V-groove runner mounted on an aluminum beam. The beam pivots near the Sharp IR sensor end and is actuated at the far end by a NEMA 17 stepper motor through a linkage. The IR sensor measures ball distance from its sensing face, while an AS5600 magnetic sensor measures beam angle on the actuator side.

2.2 Coordinates and Sign Convention

The generalized coordinates are the ball position r measured from the pivot and the beam angle θ measured from the horizontal. The analytical sign convention from `modeling.md` is:

- $r > 0$ means the ball is farther from the pivot and closer to the motor end.
- $\theta > 0$ means the motor side of the beam is up.
- Under this analytical convention, positive θ makes gravity pull the ball toward the pivot, so r decreases.

The firmware uses a calibrated runtime angle `theta_rel_deg` whose sign is the opposite of the analytical small-angle model. That distinction is kept explicit in the control chapters so the mechanics remain clean while the implementation record stays faithful to the hardware.

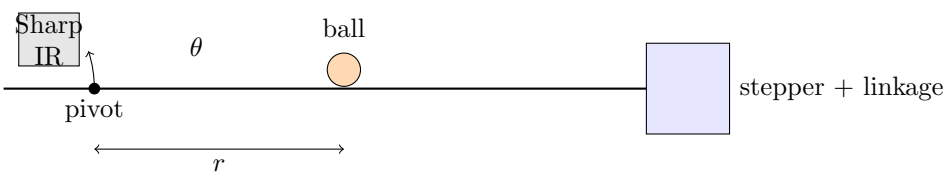

Figure 2.1: Simplified ball and beam geometry used throughout the report. Positive ball position is measured away from the pivot; positive analytical beam angle raises the motor side.

Table 2.1: Measured hardware parameters taken from the canonical modeling record.

Quantity	Value	Notes
Ball mass	2.800 g	Standard 40.000 mm ping-pong ball
Ball diameter	40.000 mm	Hollow thin-shell approximation
Bare beam mass	33.200 g	Aluminum extrusion over 263.000 mm span
Sensor plus wires mass	5.300 g	Pivot-mounted Sharp assembly
Total beam assembly mass	38.500 g	Beam, sensor, and wiring lumped together
Beam length	263.000 mm	Clevis to clevis
Sensor face from pivot	40.300 mm	Reference for the distance measurement
Nearest usable ball distance from sensor	44.000 mm	Tape stop prevents foldback region
Farthest usable ball distance from sensor	126.600 mm	Far stop in the tested runner window
Stepper resolution	3200.000 microsteps/rev	200.000 full steps with 1/16 microstepping
Rolling viscous damping	0.013 N s/m	Pre-identification estimate
Pivot viscous damping	0.006 N m s	Pre-identification estimate

2.3 Measured Hardware Parameters

Table 2.1 summarizes the measured geometric and inertial values used in the model.

The corresponding evaluated SI parameters used later are:

- ball radius $R = 0.020\,\mathrm{m}$,
- ball inertia $I = \frac{2}{3}mR^2 = 7.467 \times 10^{-7}\,\mathrm{kg\,m^2}$,
- beam assembly mass $M_b = 0.039\,\mathrm{kg}$,
- beam center of mass $L_{\mathrm{com}} = 0.118\,\mathrm{m}$,
- beam inertia about the pivot $J = 7.715 \times 10^{-4}\,\mathrm{kg\,m^2}$, and
- nominal center operating point $r_0 = 0.126\,\mathrm{m}$.

2.4 Controller-Relevant Coordinate Translation

The firmware record in `modeling.md` and `src/main.cpp` defines:

$$\theta_{\mathrm{cal,deg}} = 0.07666806 \cdot \theta_{\mathrm{AS5600,deg}} - 23.28443907 \tag{2.1}$$

$$\theta_{\mathrm{rel,deg}} = \theta_{\mathrm{cal,deg}} - \theta_{\mathrm{balance,deg}} \tag{2.2}$$

$$x_{\mathrm{cm}} = d_{\mathrm{filt,cm}} - d_{\mathrm{setpoint,cm}} \tag{2.3}$$

with the empirically verified meanings:

- $x_{\mathrm{cm}} > 0$ means the ball is toward the motor end,

- manual command `A+` rolls the ball toward the motor end, and
- near the operating point, $\theta_{\mathrm{rel,deg}} \approx -\theta_{\mathrm{phys,deg}}$.

This split between analytical and firmware coordinates is central to interpreting both the model and the telemetry correctly.

Chapter 3

Nonlinear Modeling and Derivation

3.1 Kinematics

With the pivot taken as the origin of an inertial frame, the ball center coordinates are

$$x_I = r\cos\theta \tag{3.1}$$
$$y_I = r\sin\theta \tag{3.2}$$

where r is the distance of the ball along the beam and θ is the beam angle from horizontal. Differentiating gives the inertial velocity components:

$$\dot{x}_I = \dot{r}\cos\theta - r\dot{\theta}\sin\theta \tag{3.3}$$
$$\dot{y}_I = \dot{r}\sin\theta + r\dot{\theta}\cos\theta \tag{3.4}$$

The squared speed simplifies to the standard polar-coordinate result:

$$v^2 = \dot{x}_I^2 + \dot{y}_I^2 \tag{3.5}$$
$$= \left(\dot{r}\cos\theta - r\dot{\theta}\sin\theta\right)^2 + \left(\dot{r}\sin\theta + r\dot{\theta}\cos\theta\right)^2 \tag{3.6}$$
$$= \dot{r}^2 + r^2\dot{\theta}^2 \tag{3.7}$$

The ball is modeled as rolling without slip along the beam, so its spin rate is

$$\omega_{\text{ball}} = \frac{\dot{r}}{R} \tag{3.8}$$

This removes the ball spin as an independent degree of freedom.

3.2 Energy Terms

3.2.1 Ball Translational and Rotational Kinetic Energy

The translational kinetic energy is

$$T_{\text{trans}} = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) \tag{3.9}$$

Using the thin-shell inertia $I = \frac{2}{3}mR^2$ and the rolling constraint, the rotational kinetic energy becomes

$$\begin{aligned} T_{\text{rot}} &= \frac{1}{2}I\omega_{\text{ball}}^2 && (3.10)\\ &= \frac{1}{2}\left(\frac{2}{3}mR^2\right)\left(\frac{\dot{r}}{R}\right)^2 && (3.11)\\ &= \frac{1}{3}m\dot{r}^2 && (3.12) \end{aligned}$$

3.2.2 Beam Kinetic Energy

The beam assembly rotates as a rigid body about the pivot:

$$T_{\text{beam}} = \frac{1}{2}J\dot{\theta}^2 \tag{3.13}$$

3.2.3 Total Kinetic Energy

Summing all contributions gives

$$\begin{aligned} T &= T_{\text{trans}} + T_{\text{rot}} + T_{\text{beam}} && (3.14)\\ &= \frac{5}{6}m\dot{r}^2 + \frac{1}{2}\left(mr^2 + J\right)\dot{\theta}^2 && (3.15) \end{aligned}$$

The $\frac{5}{6}$ factor on $\dot{r}^2$ is the characteristic effective-mass term of the ball and beam plant.

3.2.4 Potential Energy

Taking the pivot height as the reference, the ball and beam potential energies are

$$\begin{aligned} V_{\text{ball}} &= mgr\sin\theta && (3.16)\\ V_{\text{beam}} &= M_b g L_{\text{com}}\sin\theta && (3.17) \end{aligned}$$

so the total potential energy is

$$V = (mr + M_b L_{\text{com}})g\sin\theta \tag{3.18}$$

3.2.5 Lagrangian

The Lagrangian is therefore

$$\mathcal{L} = \frac{5}{6}m\dot{r}^2 + \frac{1}{2}\left(mr^2 + J\right)\dot{\theta}^2 - (mr + M_b L_{\text{com}})g\sin\theta \tag{3.19}$$

3.3 Euler–Lagrange Equations

The generalized forces include viscous rolling damping and actuator torque:

$$Q_r = -b_x \dot{r} \tag{3.20}$$
$$Q_\theta = \tau - b_\theta \dot{\theta} \tag{3.21}$$

3.3.1 Ball Equation

For the r coordinate,

$$\frac{\partial \mathcal{L}}{\partial \dot{r}} = \frac{5}{3} m \dot{r} \tag{3.22}$$
$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{r}}\right) = \frac{5}{3} m \ddot{r} \tag{3.23}$$
$$\frac{\partial \mathcal{L}}{\partial r} = m r \dot{\theta}^2 - m g \sin\theta \tag{3.24}$$

Applying the Euler–Lagrange equation yields

$$\frac{5}{3} m \ddot{r} + b_x \dot{r} - m r \dot{\theta}^2 + m g \sin\theta = 0 \tag{3.25}$$

The terms represent the effective rolling inertia, viscous damping, centrifugal coupling from beam rotation, and gravitational acceleration along the beam.

3.3.2 Beam Equation

For the θ coordinate,

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = (m r^2 + J)\dot{\theta} \tag{3.26}$$
$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) = (m r^2 + J)\ddot{\theta} + 2 m r \dot{r} \dot{\theta} \tag{3.27}$$
$$\frac{\partial \mathcal{L}}{\partial \theta} = -(m r + M_b L_{\text{com}}) g \cos\theta \tag{3.28}$$

Hence

$$(J + m r^2)\ddot{\theta} + 2 m r \dot{r} \dot{\theta} + b_\theta \dot{\theta} + (m r + M_b L_{\text{com}}) g \cos\theta = \tau \tag{3.29}$$

The beam equation captures inertia variation with ball position, Coriolis coupling, pivot damping, gravity, and motor torque.

3.4 Explicit Nonlinear State Model

Defining the effective rolling mass

$$m_e = \frac{5}{3}m = 0.005\,\mathrm{kg} \tag{3.30}$$

the ball equation can be written explicitly as

$$\ddot{r} = \frac{mr\dot{\theta}^2}{m_e} - \frac{mg}{m_e}\sin\theta - \frac{b_x}{m_e}\dot{r} \tag{3.31}$$

and the beam equation becomes

$$\ddot{\theta} = \frac{\tau - 2mr\dot{r}\dot{\theta} - b_\theta\dot{\theta} - (mr + M_b L_{\mathrm{com}})g\cos\theta}{J + mr^2} \tag{3.32}$$

Together, these equations form the coupled nonlinear plant with state $[r, \dot{r}, \theta, \dot{\theta}]^\mathsf{T}$. They are the source model used for the linearized design discussion and for interpreting the runtime controller structure.

Chapter 4

Linearization and Numerical Specialization

4.1 Operating Point

The control design operating point is the horizontal beam equilibrium with the ball at rest at the center of the usable sensing window:

$$r = r_0 = 0.126\,\mathrm{m}, \qquad \theta = 0, \qquad \dot{r} = 0, \qquad \dot{\theta} = 0 \tag{4.1}$$

At this operating point the actuator must balance gravity with

$$\tau_0 = (mr_0 + M_b L_{\mathrm{com}})g \tag{4.2}$$

In the implemented cascaded controller, this equilibrium torque is handled implicitly through the beam-angle servo and balance-zero procedure.

4.2 Perturbation Variables

Define small deviations from equilibrium by

$$r = r_0 + x \tag{4.3}$$
$$\theta = \delta \tag{4.4}$$
$$\dot{r} = \dot{x} \tag{4.5}$$
$$\dot{\theta} = \dot{\delta} \tag{4.6}$$

Substituting these expressions into the nonlinear equations and neglecting second-order products gives the linearized model.

4.3 Linearized Ball Dynamics

Starting from (3.25):

$$\frac{5}{3}m\ddot{x} + b_x\dot{x} + mg\delta = 0 \tag{4.7}$$

Dividing through by the effective rolling mass $m_e = \frac{5}{3}m$ gives

$$\ddot{x} + \beta\dot{x} = -\alpha g\delta \tag{4.8}$$

with

$$\alpha = \frac{m}{m_e} = \frac{3}{5} = 0.6 \tag{4.9}$$

$$\beta = \frac{b_x}{m_e} = \frac{0.0125}{0.0046667} = 2.679\,\mathrm{s}^{-1} \tag{4.10}$$

Numerically, the ball-position plant becomes

$$\ddot{x} + 2.6786\dot{x} = -5.886\,\delta \tag{4.11}$$

or in transfer-function form,

$$\frac{X(s)}{\Theta(s)} = \frac{-5.886}{s^2 + 2.6786s} = \frac{-5.886}{s(s + 2.6786)} \tag{4.12}$$

4.4 Linearized Beam Dynamics

Linearizing (3.29) gives

$$(J + mr_0^2)\ddot{\delta} + b_\theta\dot{\delta} + (mr_0 + M_bL_{\mathrm{com}})g\delta = \tau - \tau_0 \tag{4.13}$$

Substituting the evaluated parameters yields

$$J_{\mathrm{total}}(r_0) = J + mr_0^2 = 8.157 \times 10^{-4}\,\mathrm{kg\,m^2} \tag{4.14}$$

$$(mr_0 + M_bL_{\mathrm{com}})g = 0.048\,\mathrm{N\,m/rad} \tag{4.15}$$

Ignoring damping for an estimate, the beam mode has natural frequency

$$\omega_{\mathrm{beam}} = \sqrt{\frac{0.04804}{8.157 \times 10^{-4}}} = 7.670\,\mathrm{rad/s} \approx 1.220\,\mathrm{Hz} \tag{4.16}$$

4.5 Specialized Parameters

Table 4.1 collects the key values used for control interpretation.

Table 4.1: Numerically specialized parameters around the nominal operating point.

Symbol / quantity	Value	Meaning
m_e	$0.005\,\mathrm{kg}$	Effective rolling mass of the hollow ball
α	0.6	Ball-to-effective-mass ratio
β	$2.679\,\mathrm{s}^{-1}$	Linearized viscous coefficient in the ball equation
αg	$5.886\,\mathrm{m/s}^2$	Beam-angle to ball-acceleration gain
$J_{\mathrm{total}}(r_0)$	$8.157 \times 10^{-4}\,\mathrm{kg\,m}^2$	Total beam-side rotational inertia at center position
$(mr_0 + M_b L_{\mathrm{com}})g$	$0.048\,\mathrm{N\,m/rad}$	Linearized gravitational restoring torque coefficient
ω_{beam}	$7.670\,\mathrm{rad/s}$	Approximate beam natural frequency without extra actuator dynamics

4.6 Implication for Controller Structure

The linearized ball equation has an integrator and a stable real pole, while the beam subsystem is comparatively faster and directly actuated. That separation motivates the implemented cascaded structure:

- an inner loop regulates beam angle using the AS5600 measurement and stepper actuation, and
- an outer loop commands beam angle from ball-position error, velocity, and a constrained integral term.

The actual firmware is more detailed than the ideal linear model. It retains quantization, deadband, validity gating, recovery logic, and empirical limits, which are described in the next chapter.

Chapter 5

Firmware and Control Implementation

5.1 Runtime State Definitions

The firmware quantities recorded in the April 11 telemetry CSV are expressed in centimeters, degrees, and seconds. The most important runtime states are:

$$d_{\text{raw,cm}} : \text{ instantaneous Sharp distance estimate} \tag{5.1}$$
$$d_{\text{filt,cm}} : \text{ filtered distance used for control} \tag{5.2}$$
$$x_{\text{cm}} = d_{\text{filt,cm}} - d_{\text{setpoint,cm}} \tag{5.3}$$
$$\dot{x}_{\text{cm/s}} : \text{ filtered ball-velocity estimate} \tag{5.4}$$
$$\theta_{\text{cal,deg}} : \text{ calibrated beam angle before balance zeroing} \tag{5.5}$$
$$\theta_{\text{rel,deg}} : \theta_{\text{cal,deg}} - \theta_{\text{balance,deg}} \tag{5.6}$$
$$\dot{\theta}_{\text{deg/s}} : \text{ filtered beam-angle rate estimate} \tag{5.7}$$
$$\xi_{\text{cm.s}} : \text{ outer-loop integral state} \tag{5.8}$$

5.2 Sensing and Signal Conditioning

5.2.1 Sharp Distance Sensor

At each 40.000 ms cycle the controller averages eight ADC samples and applies the inverse-voltage fit

$$d_{\text{raw,cm}} = \frac{12.25}{V_{\text{sharp}}} - 0.62 \tag{5.9}$$

with the validity limits

$$V_{\text{sharp}} \geq 0.080\,\text{V}, \qquad 4.400\,\text{cm} \leq d_{\text{raw,cm}} \leq 12.660\,\text{cm} \tag{5.10}$$

Accepted samples pass through a three-sample median and two exponential filters:

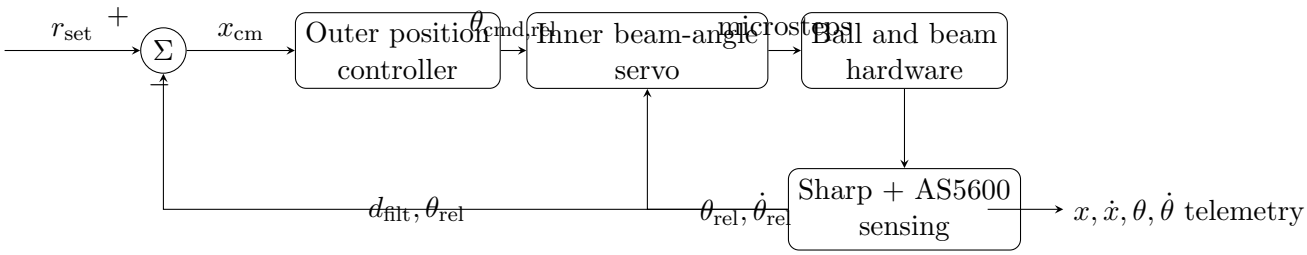

Figure 5.1: Implementation-oriented cascaded control structure used by the April 11 telemetry run.

$$m[k] = \text{median of last three valid distance samples} \tag{5.11}$$

$$d_{\text{filt}}[k] = 0.25\, m[k] + 0.75\, d_{\text{filt}}[k-1] \tag{5.12}$$

$$d_{\text{vel}}[k] = 0.60\, m[k] + 0.40\, d_{\text{vel}}[k-1] \tag{5.13}$$

$$d_{\text{rate}}[k] = \frac{d_{\text{vel}}[k] - d_{\text{vel}}[k-1]}{0.04} \tag{5.14}$$

The ball-velocity estimate is then formed with adaptive smoothing so the near-sensor side receives extra filtering.

5.2.2 AS5600 Beam Angle

The beam-angle conversion used online is the affine fit

$$\theta_{\text{cal,deg}} = 0.07666806 \cdot \theta_{\text{AS5600,deg}} - 23.28443907 \tag{5.15}$$

The balance-zero command sets $\theta_{\text{balance,deg}}$ so that $\theta_{\text{rel,deg}} = 0$ corresponds to the chosen control reference rather than the raw iPhone-level zero from the calibration session.

5.3 Mode Structure

The staged firmware in `src/main.cpp` provides:

- **M0**: telemetry only, no control output,
- **M1**: manual beam-angle hold with the inner loop only, and
- **M2**: cascaded ball-position control.

The driver starts disabled at boot. The operator enables it explicitly, captures the balance reference with the `Z` command, and then enters the cascaded mode. This staged flow reduces risk while tuning the hardware.

5.4 Implemented Cascade

5.4.1 Inner Beam-Angle Servo

The commanded beam angle is first limited to the calibrated soft window $[-0.90,\ 3.30]$ degrees around the balance reference. The beam-angle error is

$$e_{\theta,\mathrm{deg}} = \theta_{\mathrm{cmd,cal,deg}} - \theta_{\mathrm{cal,deg}} \tag{5.16}$$

If both the angle error and beam-rate magnitude are inside the deadband, $|e_\theta| \leq 0.040\,\mathrm{deg}$ and $|\dot{\theta}_{\mathrm{deg/s}}| \leq 0.600\,\mathrm{deg/s}$, the controller outputs zero steps. Otherwise the discrete servo computes

$$\mathrm{raw_steps}[k] = -\left(0.70\, e_{\theta,\mathrm{deg}}[k]\right) \cdot 115.9399 + \mathrm{residual}[k-1] \tag{5.17}$$

and rounds the result to an integer command clipped to ± 40 microsteps per cycle.

5.4.2 Outer Ball-Position Law

The outer loop clamps the derivative states to

$$|\dot{x}_{\mathrm{cm/s}}| \leq 4.500\,\mathrm{cm/s}, \qquad |\dot{\theta}_{\mathrm{deg/s}}| \leq 4.000\,\mathrm{deg/s} \tag{5.18}$$

and then applies the tuned state-feedback law

$$\theta_{\mathrm{fs,cmd,deg}} = g_{\mathrm{outer_sign}} \left(k_x x_{\mathrm{cm}} + k_v \dot{x}_{\mathrm{cm/s}} + k_t \theta_{\mathrm{rel,deg}} + k_\omega \dot{\theta}_{\mathrm{deg/s}} + i_{\mathrm{term,deg}} \right) \tag{5.19}$$

with the April 11 defaults

$$g_{\mathrm{outer_sign}} = -1 \tag{5.20}$$
$$k_x = 0.28 \tag{5.21}$$
$$k_v = 0.20 \tag{5.22}$$
$$k_t = 0.00 \tag{5.23}$$
$$k_\omega = 0.00 \tag{5.24}$$
$$k_i = 0.020 \tag{5.25}$$

The integral state is constrained to the near-target capture region and clipped so that $|i_{\mathrm{term,deg}}| \leq 0.180\,\mathrm{deg}$.

5.4.3 Recovery and Trim Logic

The implementation retains three important practical features:

- a **recovery floor** that enforces a minimum corrective beam command when the ball appears stalled away from the setpoint,
- a **zero-trim estimator** that logs a slow local balance-angle estimate under tight near-equilibrium conditions, and
- a **position-trim map** that can learn a local hold-angle bias across the usable setpoint range.

No learned position-trim feedforward was active during the April 11 run, so the telemetry shows `theta_trim_ff_deg = 0` for every row.

5.5 Implementation Constants

Table 5.1 compresses the runtime constants most relevant to reproducing the telemetry run.

Table 5.1: Key runtime constants from the firmware implementation in `src/main.cpp`.

Group	Constants	Value	Meaning
Timing	`LOOP_MS`, `DT`	40.000 ms, 0.040 s	Main control cadence
Distance window	`D_MIN_CM`, `D_MAX_CM`, `D_SETPOINT_DEFAULT_CM`	4.400 cm, 12.660 cm, 8.530 cm	Allowed sensing and default center setpoint
Sharp fit	`SHARP_FIT_K_V_CM`, `SHARP_FIT_OFFSET_CM`	12.25, -0.62	Inverse-voltage distance calibration
Angle calibration	`THETA_SLOPE_DEG_PER_AS_DEG`, `THETA_OFFSET_DEG`	0.07666806, -23.28443907	Online AS5600 to beam-angle affine fit
Angle limits	`THETA_CAL_MIN_DEG`, `THETA_CAL_MAX_DEG`	-0.70, 3.10	Measured calibrated beam-angle range
Stepper conversion	`STEPS_PER_REV`, `STEPS_PER_BEAM_DEG`	3200, 115.9399	Empirical beam-angle to microstep conversion
Inner loop	`INNER_KP_THETA`, `INNER_KD_THETA`, `INNER_MAX_STEP_DELTA`	0.70, 0.00, 40	Discrete beam-angle servo gains and step clamp
Outer loop	`OUTER_KX_DEFAULT`, `OUTER_KV_DEFAULT`, `OUTER_KI_DEFAULT`	0.28, 0.20, 0.020	Default position-feedback and integral gains
Recovery	`RECOVERY_ENTER_X_CM`, `RECOVERY_FLOOR_DEFAULT_DEG`, `RECOVERY_FLOOR_MAX_DEG`	1.200 cm, 0.700 deg, 1.100 deg	Stall detection and minimum corrective angle floor
Staircase	`STAIRCASE_STAGE_MS`, `STAIRCASE_FAR_SETPOINT_CM`, `STAIRCASE_CENTER_SETPOINT_CM`, `STAIRCASE_NEAR_SETPOINT_CM`	20 000.000 ms, 12.16, 8.53, 4.90 cm	Built-in experiment sequence used in the telemetry log

Chapter 6

Beam-Angle Calibration Analysis

This chapter analyzes the combined forward and reverse calibration data stored in `calibration_logs/combined_20260407_forward_reverse/combined_cleaned_calibration.csv`. Those data are used here as the direct source for beam-angle calibration figures and hysteresis statistics.

6.1 Scope of the Calibration Data

The combined CSV merges a forward sweep and an interpolated reverse sweep onto a common AS5600 angle grid. Its role in this report is to show the measured static beam-angle relationship and the amount of direction-dependent hysteresis. It is not the runtime controller source-of-truth by itself.

The implemented controller still uses the same-session affine calibration documented in `modeling.md` and `src/main.cpp`:

$$\theta_{\text{cal,deg}} = 0.07666806 \cdot \theta_{\text{AS5600,deg}} - 23.28443907 \tag{6.1}$$

That distinction matters because the combined forward/reverse dataset shows larger direction-dependent offset than the same-session calibration chosen for closed-loop control.

6.2 Calibration Summary

The combined CSV contains 112 grid points. The observed reverse minus forward hysteresis is summarized by:

- mean hysteresis: 0.418 degrees,
- RMS hysteresis: 0.422 degrees, and
- max hysteresis: 0.600 degrees.

6.3 Forward/Reverse Curve

Figure 6.1 shows that the relationship is approximately affine over the operating window but not perfectly single-valued when travel direction is considered. The combined dataset therefore

Table 6.1: Summary statistics generated directly from the combined calibration CSV.

Metric	Value
Grid points	112
AS5600 range [deg]	290.391 to 340.928
Forward theta range [deg]	-1.300 to 2.400
Reverse theta range [deg]	-0.900 to 2.800
Mean hysteresis [deg]	0.418
RMS hysteresis [deg]	0.422
Max hysteresis [deg]	0.600
Forward affine fit	0.07643 x + -23.62397
Reverse affine fit	0.07435 x + -22.54843

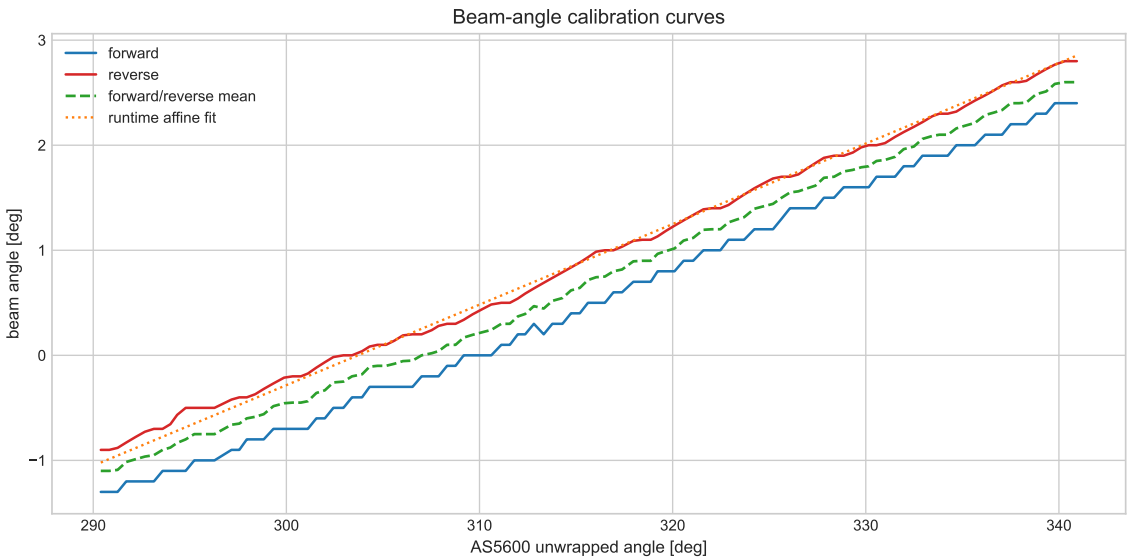

Figure 6.1: Forward and reverse beam-angle calibration curves generated directly from the combined calibration CSV. The overlaid affine line is the runtime mapping used in the controller.

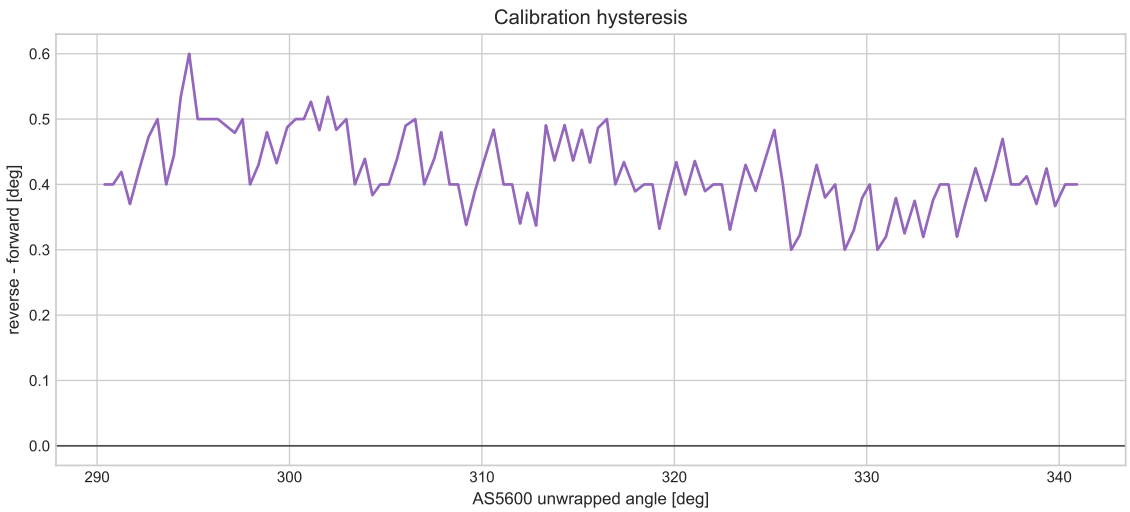

Figure 6.2: Direction-dependent hysteresis, plotted as reverse minus forward beam angle versus unwrapped AS5600 angle.

serves best as an offline characterization tool.

6.4 Hysteresis Plot

The hysteresis remains positive across the combined dataset, with a magnitude large enough to justify separating the thesis-facing static characterization from the simpler runtime affine map used for controller execution.

Chapter 7

Experiment Results

7.1 Telemetry Mapping and Analysis Assumptions

The results in this chapter come from `telemetry_logs/20260411_142117/telemetry.csv`. The analysis script uses the logged columns `t_ms`, `setpoint_cm`, `d_filt_cm`, `x_cm`, `x_dot_cm_s`, `theta_rel_deg`, `theta_dot_deg_s`, `theta_cmd_rel_deg`, `mode`, `valid`, `driver_enabled`, `recovery_active`, `step_pos`, and `step_delta`.

Only rows with `mode == 2` and `driver_enabled == 1` are treated as closed-loop experiment data. Position-derived metrics also honor the logged `valid` flag because the far-step segment contains intermittent invalid Sharp samples.

For figure readability, the position and error plots use a display-only five-sample centered rolling mean of the logged position signal, and short interior invalid gaps are bridged only in the drawn trace. The summary tables and reported metrics still use the raw valid samples from the CSV.

7.2 Run Summary

The full log contains 364 rows spanning 72.601 seconds, with 350 rows in closed-loop cascade mode. The run contains 43 invalid sensor rows and 9 rows with recovery active. The maximum observed absolute tracking error was 3.712 cm, while the maximum measured beam-angle magnitude and command magnitude were 1.543 deg and 1.344 deg respectively.

7.3 Overview Plot

The staircase sequence consists of an initial center hold, a far-side step, a return to center, and a near-side step. The local time axes make the settled portions legible without altering the underlying data. Validity losses are concentrated in the far-side segment, which is why the generated summary tables separate valid sample counts from total row counts.

7.4 Ball-Position Tracking

The split transition-and-hold layout makes the experiment easier to read than a single long time axis. In particular, the far-side and near-side moves show usable late-hold tracking once the initial transport transient is separated from the rest of the phase. The center-return segment

Table 7.1: Signal mapping assumptions used by the analysis pipeline.

Telemetry column	Mapped quantity	Note
t_ms	Elapsed experiment time	Converted to seconds from the first logged sample.
setpoint_cm	Reference ball position	Controller target in Sharp distance coordinates.
d_filt_cm	Measured ball position	Filtered distance used for control and plotting.
x_cm	Tracking error	Defined in firmware as d_filt_cm - setpoint_cm.
x_dot_cm_s	Ball velocity estimate	Filtered derivative used by the outer loop.
theta_rel_deg	Measured beam angle	Runtime beam-angle coordinate after balance zeroing.
theta_dot_deg_s	Beam-angle rate	Filtered derivative of theta_rel_deg.
theta_cmd_rel_deg	Final control input	Relative beam-angle command sent to the inner loop.
valid	Measurement validity	Position-derived metrics only use rows where valid = 1.
mode and driver_enabled	Closed-loop filter	Analysis keeps rows with mode = 2 and driver_enabled = 1.

Table 7.2: Run-level summary generated from the telemetry CSV.

Metric	Value
Telemetry rows	364
Closed-loop rows	350
Duration [s]	72.601
Invalid rows	43
Recovery-active rows	9
Command-saturated rows	0
Max abs tracking error [cm]	3.712
Max abs beam angle [deg]	1.543
Max abs command [deg]	1.344
Closed-loop setpoints [cm]	4.90, 8.53, 12.16

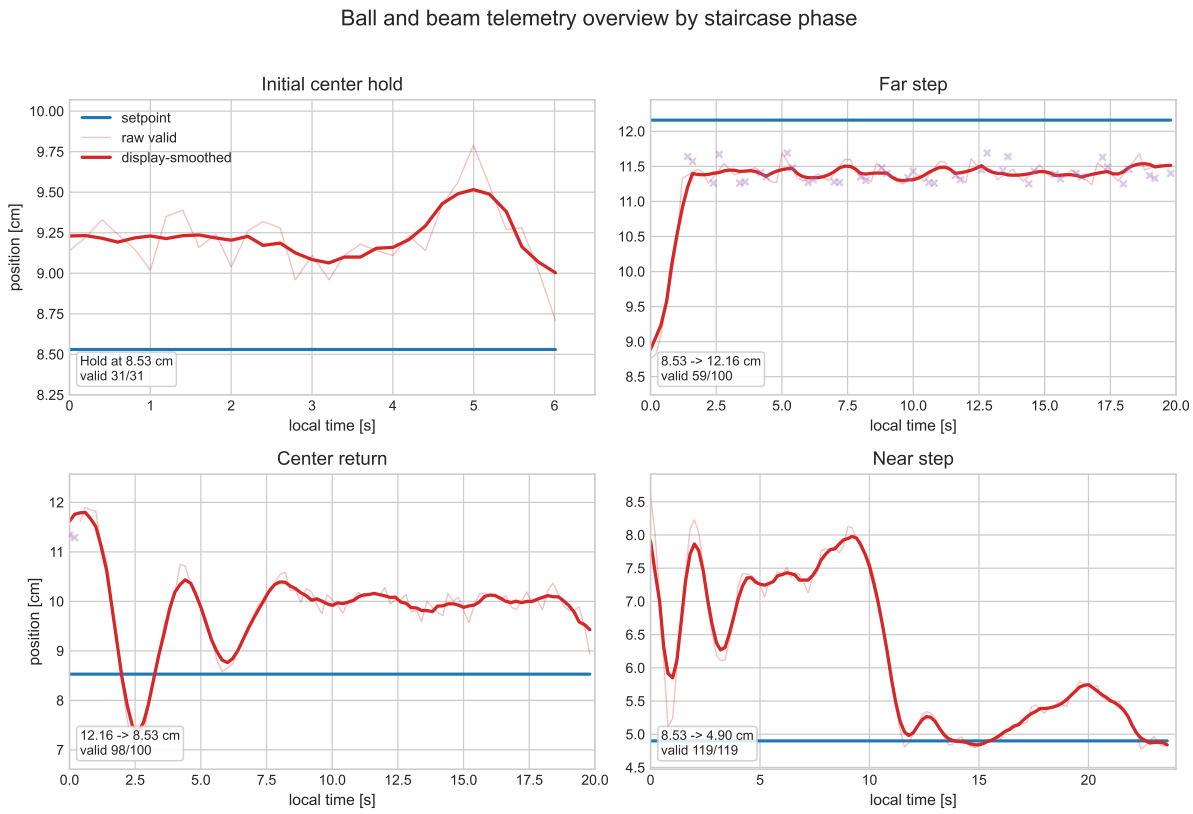

Figure 7.1: Phase-by-phase overview of the staircase experiment. Each panel uses a local time axis and its own vertical scale so the hold portions remain readable. The bold red curve is a display-smoothed version of the valid measured signal, while the faint raw trace shows the underlying sensor fluctuation.

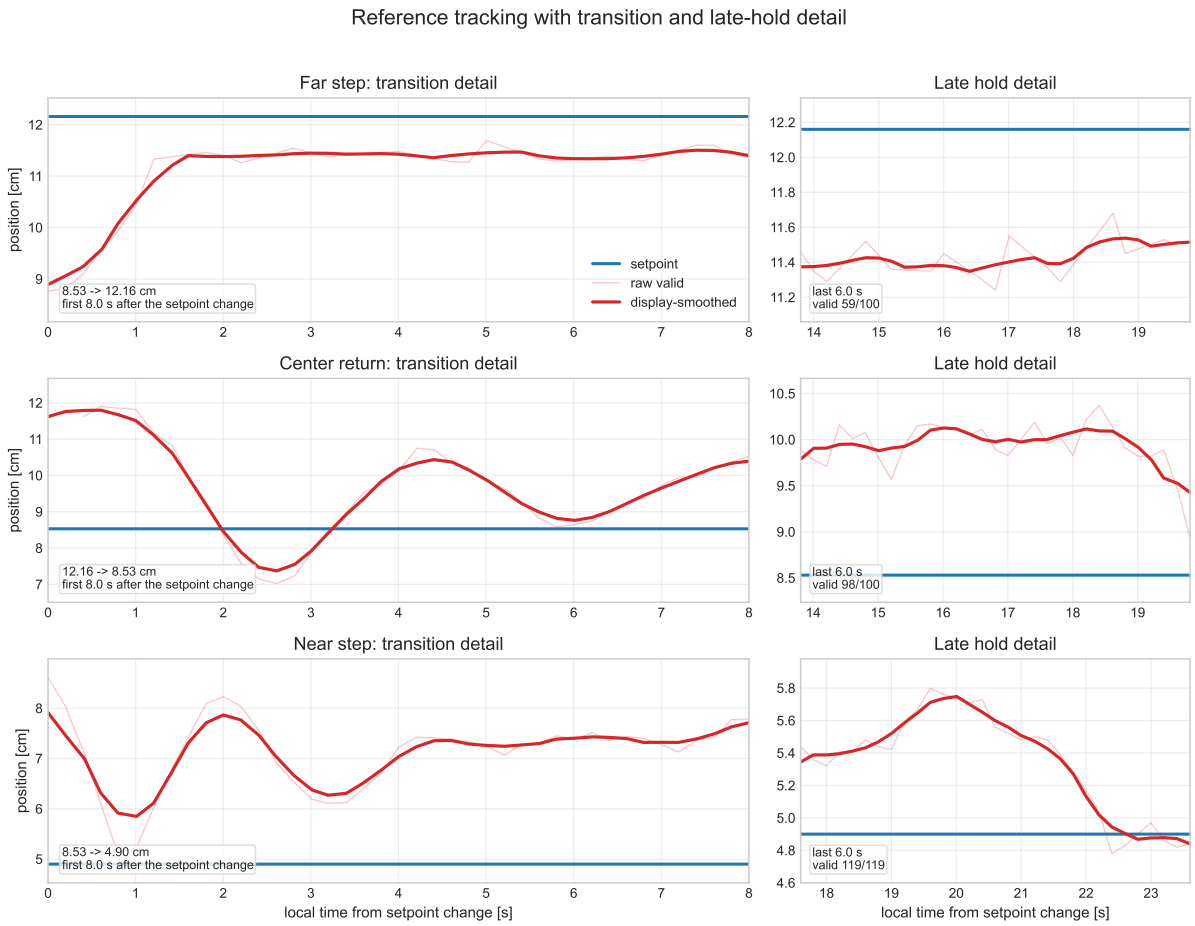

Figure 7.2: Reference tracking by commanded transition. The left column shows the first 8.000 s after each setpoint change, while the right column zooms into the late hold portion of the same segment.

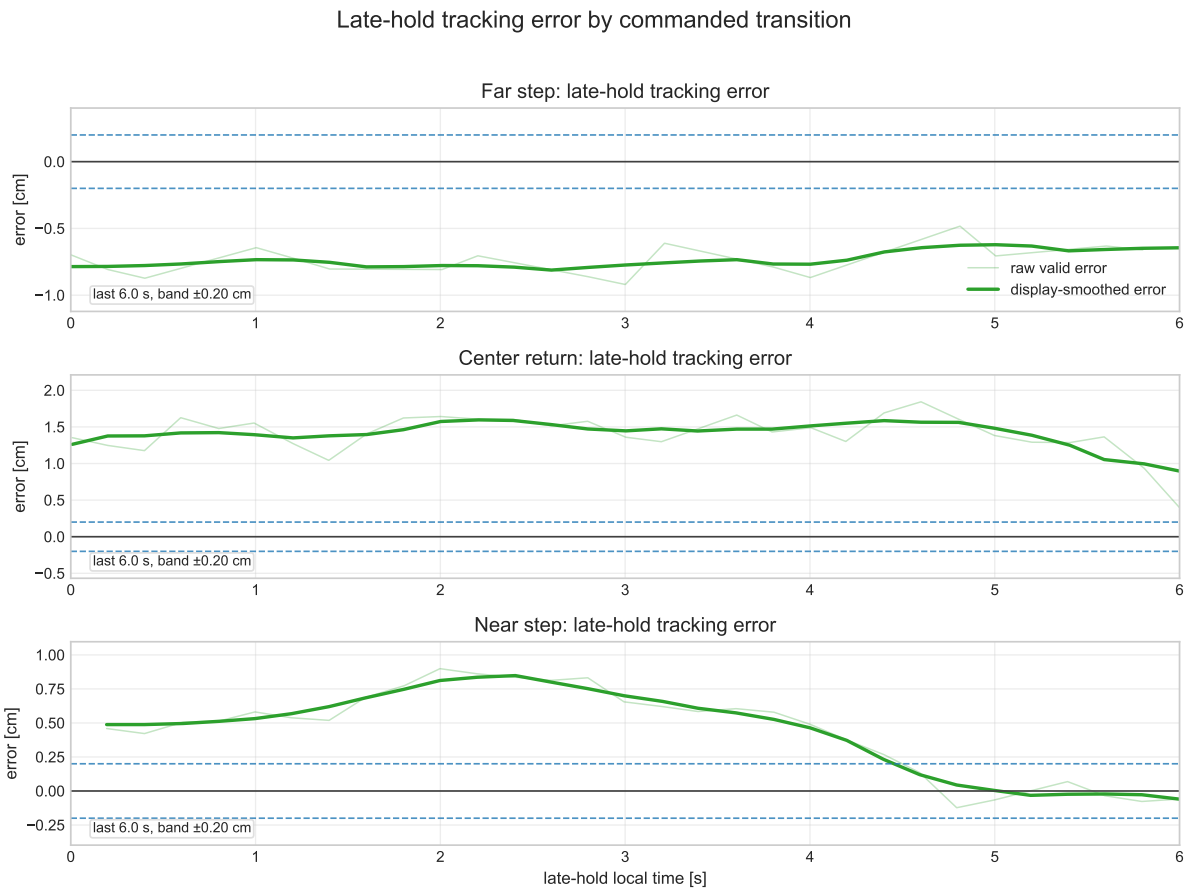

Figure 7.3: Late-hold tracking error $x_{\mathrm{cm}} = d_{\mathrm{filt,cm}} - d_{\mathrm{setpoint,cm}}$ for each commanded transition. Dashed lines show the default settling band used in the summary metrics, while the bold curve uses the same display-only smoothing as the position plots.

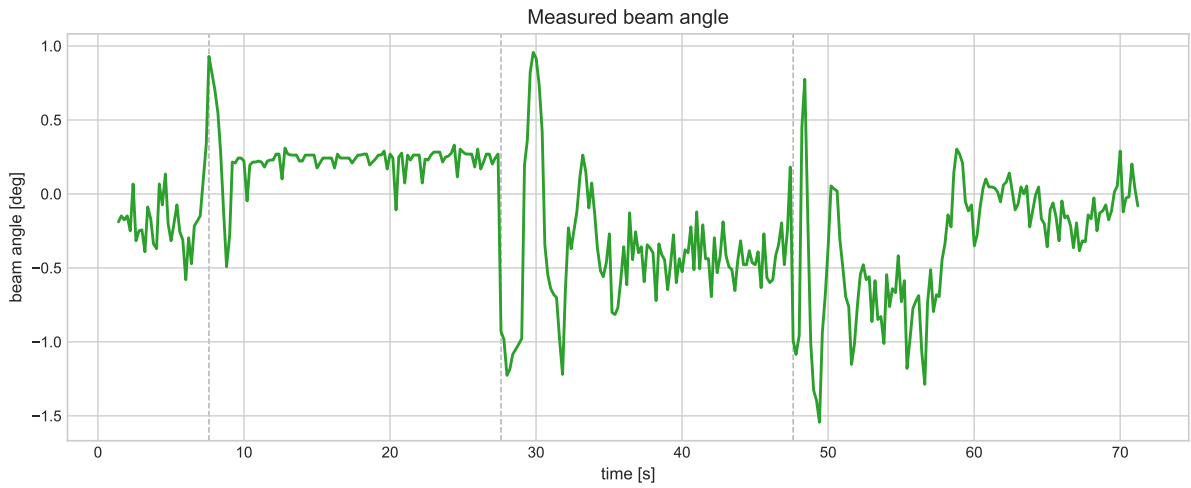

Figure 7.4: Measured beam angle versus time during the staircase experiment.

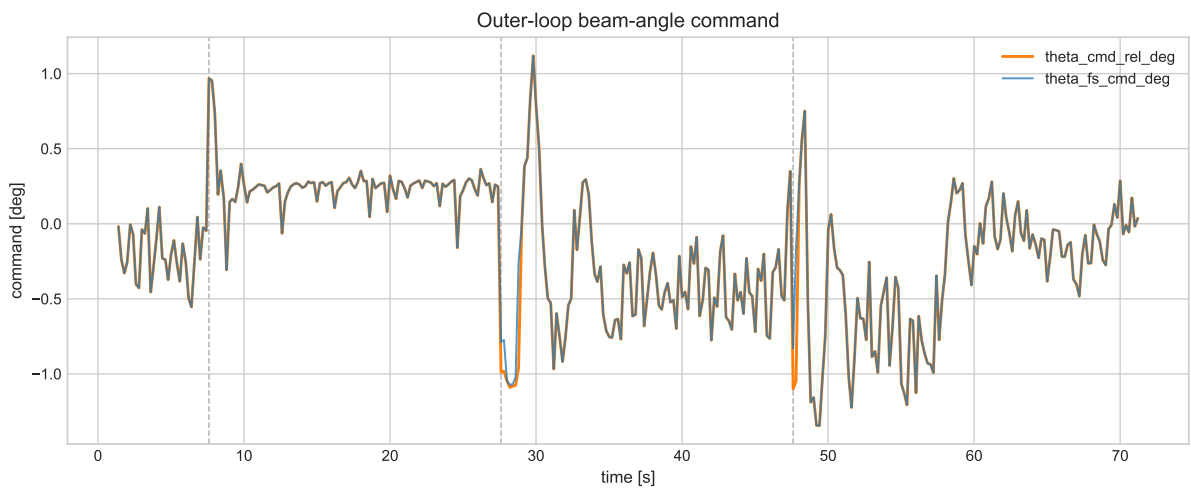

Figure 7.5: Final relative beam-angle command sent to the inner loop.

still carries the largest residual offset in the available window, so the figure presentation is clearer but not artificially optimistic.

7.5 Beam Motion and Control Effort

The logged beam-angle command remains well inside the configured 2.000 deg command cap, and the CSV shows `theta_cmd_sat = 0` for all rows. Recovery activates only briefly during large transitions, which is consistent with the firmware design intent: use recovery as a minimum-corrective-angle floor rather than as the dominant transport mechanism.

7.6 Performance Metrics

The step metrics confirm three main points:

- the far-side transition is limited by invalid distance samples and does not reach a clean settled condition in the available segment length,

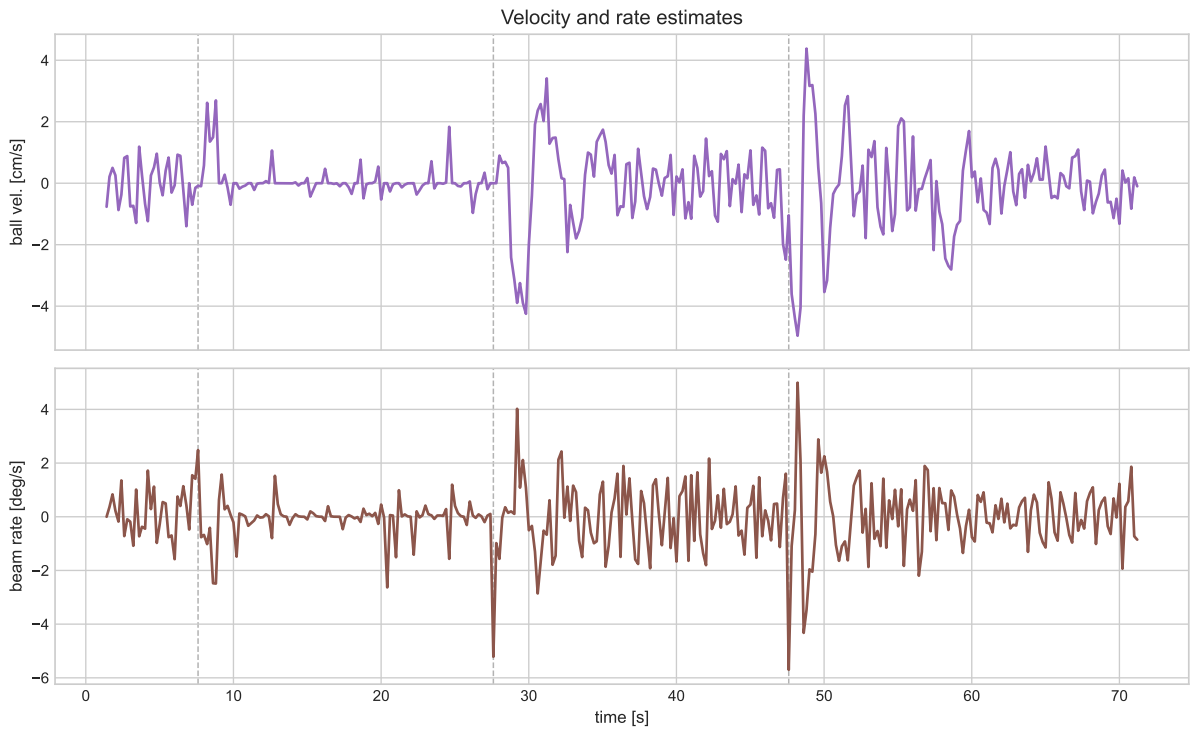

Figure 7.6: Ball velocity and beam-angle rate estimates derived from the logged telemetry.

Table 7.3: Per-step tracking metrics computed from valid closed-loop samples. Settling uses the default band $\max(0.200\,\text{cm}, 5\%$ of commanded step size). Unsupported metrics remain `N/A`.

Step	Transition	Δ [cm]	Valid	Peak $\lvert e \rvert$	RMSE	MAE	SSE	Ov.	Wrong-way	Rise / settle [s]
Far step	8.53 -¿ 12.16	3.63	59/100	3.398	1.139	0.944	-0.711	0.000	0.000	N/A / N/A
Center return	12.16 -¿ 8.53	-3.63	98/100	3.365	1.552	1.428	1.310	1.505	0.280	0.798 / N/A
Near step	8.53 -¿ 4.90	-3.63	119/119	3.712	1.692	1.257	0.049	0.123	0.000	0.600 / 22.196

- the center-return and near-side transitions achieve finite rise times but still show non-negligible residual error over the final valid window, and
- the recorded experiment is more useful as a realistic tuning baseline than as a final high-performance demonstration.

Chapter 8

Conclusion

This report system turns the repository into a self-contained ball and beam experiment record. It packages the mathematical model, runtime controller description, calibration analysis, telemetry plots, generated tables, and local build tooling into a single maintainable LaTeX tree under `report/`.

From the experiment itself, the main conclusions are:

- the nonlinear model and linearized operating-point analysis remain consistent with the staged cascaded controller architecture,
- the combined forward/reverse calibration data expose measurable hysteresis, so offline calibration analysis should remain distinct from the simpler online affine mapping used by the firmware, and
- the April 11 staircase run demonstrates functional closed-loop transport across the usable sensing window, but the response quality is still limited by sensor invalidity and residual steady-state error during larger moves.

Immediate next steps are to collect cleaner far-side telemetry, improve near/far distance validity robustness, and rerun the staircase experiment after those sensing improvements so the same report pipeline can produce a higher-confidence final results chapter.

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